

MILESTONE 1

ProSense

Modular platform for intelligent functional training monitoring based on computer vision

Group 1

António Videira · David Cálix · Diogo Ruivo · Gabriel Riquito · Tiago Oliveira



Context



Training video

The movement



Metrics

Repetitions, cadence, range of motion



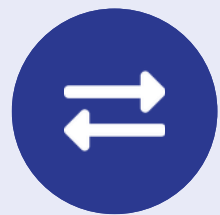
Gym management

Athletes, sessions, results

Three worlds that don't talk to each other today



is an open, modular platform that turns functional training videos into movement metrics, for coaches, athletes, physiotherapists and researchers.



Swappable pose algorithms



Exercises defined by the coach (MDL)

The algorithm



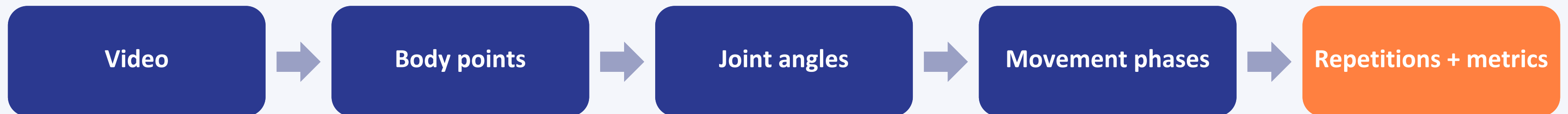
Pose estimation and joint angles across the phases of a squat.

What it is

Pose estimation plus the Movement Description Language (MDL), developed at IEETA. Each exercise is a state machine: states are movement phases, and joint angles trigger the transitions.

What it does

Finds the body points in every frame, measures joint angles, detects the phase of the movement and counts repetitions. New exercises are defined by parameters, with no model training.



Personas



Athlete

Records and submits videos, checks their metrics and progress over time, and receives feedback.



PT / Coach

Creates exercises with the MDL, schedules sessions, reviews videos with the skeleton overlay, compares performances and leaves feedback.



Physiotherapist

Tracks recovery with biomechanical metrics and defines rehabilitation. For home exercises, gets the algorithm's output on the patient's execution.



Admin

Views and creates groups (gyms, classes), links athletes and coaches, and manages users.



Researcher

Uses the platform to refine algorithms, compare pose models on the same videos and explore new uses in the future.

Expected results / Goals

1

Modular platform

Pose algorithms and robustness stages as swappable plugins.

2

Custom exercises

Coaches create and tune exercises with the MDL, with no model training.

3

Metrics and feedback

Repetitions, cadence and range of motion, with video and skeleton overlay.

All validated with real data from a functional training box.

MVP

The essentials

- ✓ Submit the video of an exercise (mobile or web app)
- ✓ Pose + MDL compute repetitions, cadence and range of motion
- ✓ Asynchronous processing through the plugin architecture
- ✓ Coaches create and configure exercises with the MDL
- ✓ Dashboard: video with skeleton, metrics and history
- ✓ Basic management of groups and users

Left for later

- 🕒 Second pose algorithm (e.g. YOLO-Pose)
- 🕒 Multiple people and multiple cameras
- 🕒 New ideas that come up along the way



The MVP can change, and easily.

This project is exploratory: along the way we might, for example, use AI to automatically detect which exercise is being performed. Features may come in and others go out. That is why we start from the essentials and a modular architecture.

The State of the Art



Yes



Partial



No, or not stated on the official website

Feature	Regybox <i>box management (PT)</i>	SugarWOD <i>workouts and community</i>	Wodify <i>full management</i>	Beyond the Whiteboard <i>logging and analysis</i>	ProSense <i>modular platform</i>
BOX AND WORKOUT MANAGEMENT					
Mobile app for athletes	✓	✓	✓	✓	✓
Workout planning (WOD)	✓	✓	✓	✓	Partial
Results logging and personal records	✓	✓	✓	✓	Partial
Class booking and payments	✓	✗	✓	✓	✗
GROUP CLASSES					
Class leaderboard	✓	✓	✓	✓	✗
Live class tracking during the session	✗	✗	✓ ¹	Partial ²	Partial ³
Class results obtained without manual logging	✗	✗	✗	✗	✓
MOVEMENT ANALYSIS					
Automatic video analysis (repetitions, cadence, range of motion)	✗	✗	✗	✗	✓
Movement criteria defined by the coach	✗	✗	✗	✗	✓
Swappable analysis algorithms	✗	✗	✗	✗	✓

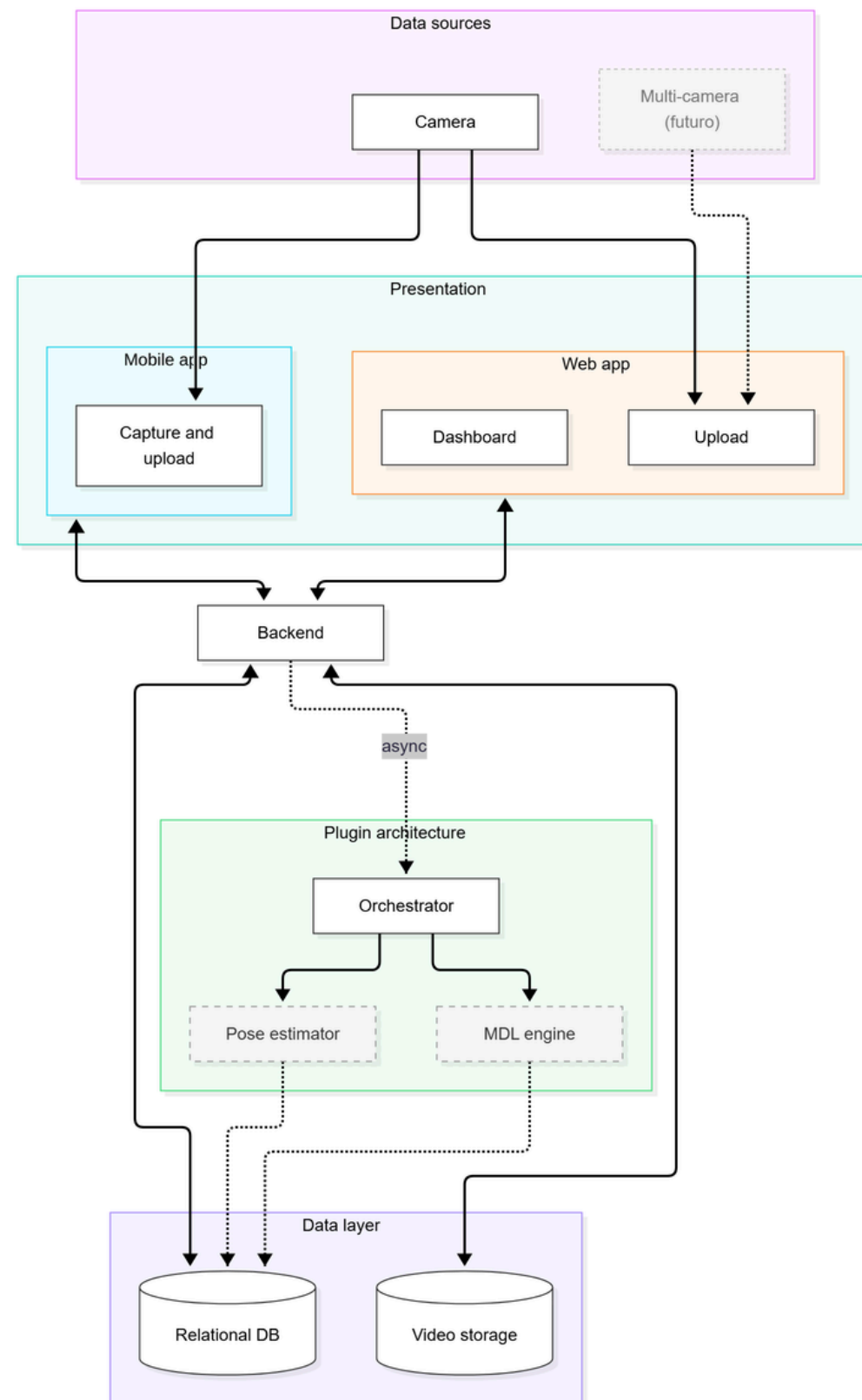
1 Wodify Pulse add-on (Myzone straps): the Kiosk+ panel shows each athlete in the class live.

2 Tempus feature: live multiplayer mode, but the athlete manually marks each change of movement.

3 In phase 1, each athlete in the class submits their own video and results are aggregated per session; simultaneous monitoring of the whole class is planned as an extension.

Sources: official websites and help centers of each product, accessed on 28 Sep 2026.

Architecture



1

Data sources

Camera in the MVP. Multi-camera setups are planned as a future extension.

2

Presentation

Mobile app (capture and upload) and web app (dashboard and video upload).

3

Backend and plugins

The backend asynchronously hands work to an orchestrator that calls swappable plugins: pose estimator and MDL engine.

4

Data layer

Relational database and video storage.

Dashed: future components or swappable plugins.

Project Calendar

Modules	Dates	Tasks list
Requirements and domain	29/09 - 27/10	1. Requirements gathering (interviews / IEETA documents) 2. Actors and use cases 3. Functional and non-functional requirements 4. State of the art (detailed)
Architecture and modeling	27/10 - 24/11	1. Deployment diagram 2. Domain model (ER) 3. UI mock-ups 4. Architecture skeleton (M2)
Core platform (backend)	24/11 - 08/12	1. Authentication and user management 2. Management of gyms, exercises and sessions (MDL) 3. Persistence layer (DAO)
Plugin architecture	24/11 - 12/12	1. Orchestrator + asynchronous task queue 2. Pose estimation plugin integration (at least 2) 3. MDL engine integration
Mobile app	24/11 - 12/12	1. Video capture and upload 2. Basic coach / athlete interface
Web app and dashboard	01/12 - 15/12	1. Video and class upload 2. Results visualization 3. Accessibility and usability review (MS3)
Testing and documentation	09/12 - 16/12	1. Unit and integration tests 2. Technical documentation and user manual 3. MVP presentation preparation (MS4)

Project Calendar: Milestones

Milestone	Date	Description
M1	29/09/2026	Lifecycle objectives and project calendar presented.
M2	24/11/2026	Architecture validated: requirements, domain model and mock-ups completed.
MS3	Dec/2026	Digital accessibility and usability review of the platform.
MS4	15-16/12/2026	MVP presented to supervisors; peer evaluation.

Highlighted: the milestone we are presenting today.

Microsite

prosense-pei.github.io/prosense-microsite



ProSense

Milestones

Meetings

Calendar

GitHub

Jira



The platform
that's **here to**
change your
training.

ProSense turns everyday training video into objective performance data — reps, tempo, range of motion and more — without a single sensor strapped to the athlete.



Thank you!



KEEP TRACK OF OUR WORK!



Gabriel Riquito



António Videira



David Cálix



Tiago Oliveira



Diogo Ruivo

Supervisors: António Neves · Daniel Canedo
Collaborators: Lúcia Sousa · Pedro Santos

